

Risk Signal Engine

From breaking news and social posts to explainable risk signals and a portfolio stress test

S&P Global × CRISIL · Code to Connect Hackathon 2026 · Module B: Strategic Portfolio Stress Testing

1,325

real captured headlines and posts

1–10

explainable impact score

seconds

from headline to portfolio view

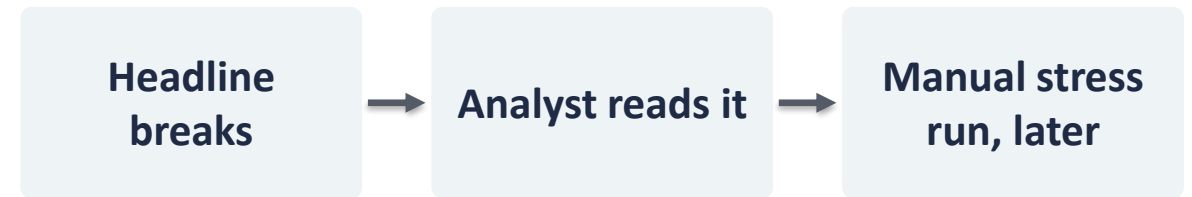
Individual submission · decision-support prototype, not investment advice

Problem and approach

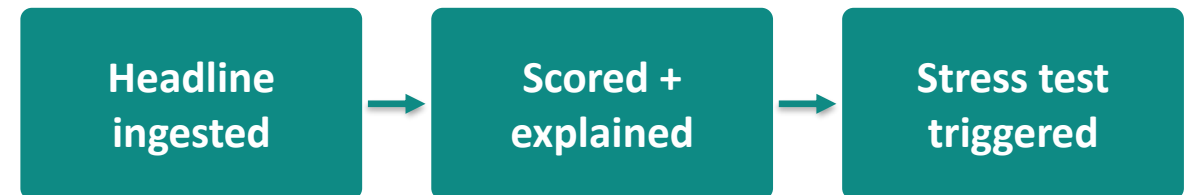
Risk shows up in text before it shows up in data

- Downgrades, probes, sanctions and rate moves break as headlines and posts
- Credit analysts and risk teams triage them by hand: slow, noisy, duplicated
- Nothing links a headline to the book's exposure until someone runs numbers
- **Approach:** an NLP risk engine scores every item (sentiment, event, impact) and an event-driven stress engine reprices the book

Today



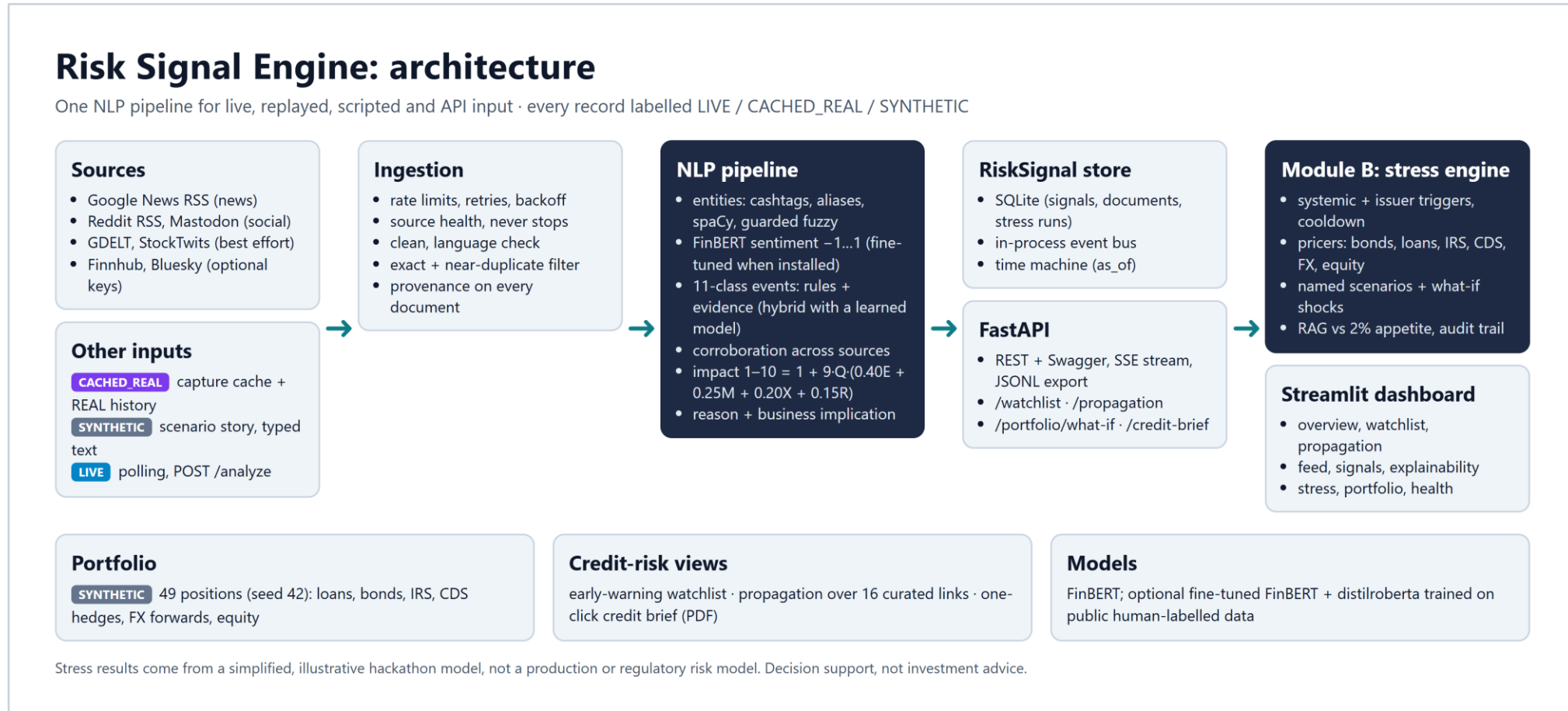
With the engine



Measured pipeline latency: median 204.7 ms per document (n = 200, CPU laptop)

System design

One NLP pipeline for live, replayed, scripted and API input



Sources → ingestion → NLP → RiskSignal store and API → stress engine and dashboard · every record labelled LIVE / CACHED_REAL / SYNTHETIC (docs/architecture.png)

Implementation highlights

Every score is explained, every weight is published

- **NLP:** entities (cashtags, aliases, spaCy, guarded fuzzy), FinBERT sentiment, 11-class event rules with evidence; optional fine-tuned models in a hybrid
- **Stress engine:** news-only systemic and issuer triggers with cooldown; pricers for bonds, loans, IRS, CDS, FX, equity; what-if sliders
- **Credit views:** watchlist, propagation over 16 curated links, one-click credit brief
- **Engineering:** one pipeline for every input, offline demo, time machine over real data, audit trail



Dashboard explainability: factor contributions for a SYNTHETIC demo signal

$$\text{Impact} = 1 + 9 \times Q \times (0.40 \cdot E + 0.25 \cdot M + 0.20 \cdot X + 0.15 \cdot R)$$

E event severity · M sentiment magnitude · X exposure · R credibility +
corroboration · Q confidence

Expert-set priors, not calibrated

Key results and metrics

Accuracy: preliminary (labels pending human review)

0.653

Sentiment

accuracy · FinBERT · n = 147

0.81

Events

accuracy · rules · n = 147

0.898

Entities

accuracy · n = 147

204.7 ms

Median latency

p95 1085.2 ms · n = 200

84 → 50

Stress runs

replay after fixes · n = 911

Preliminary (labels pending human review): labels drafted by an AI assistant, which also wrote the event rules. Treat as a smoke test, not a benchmark.

Scenario demo (simulated)

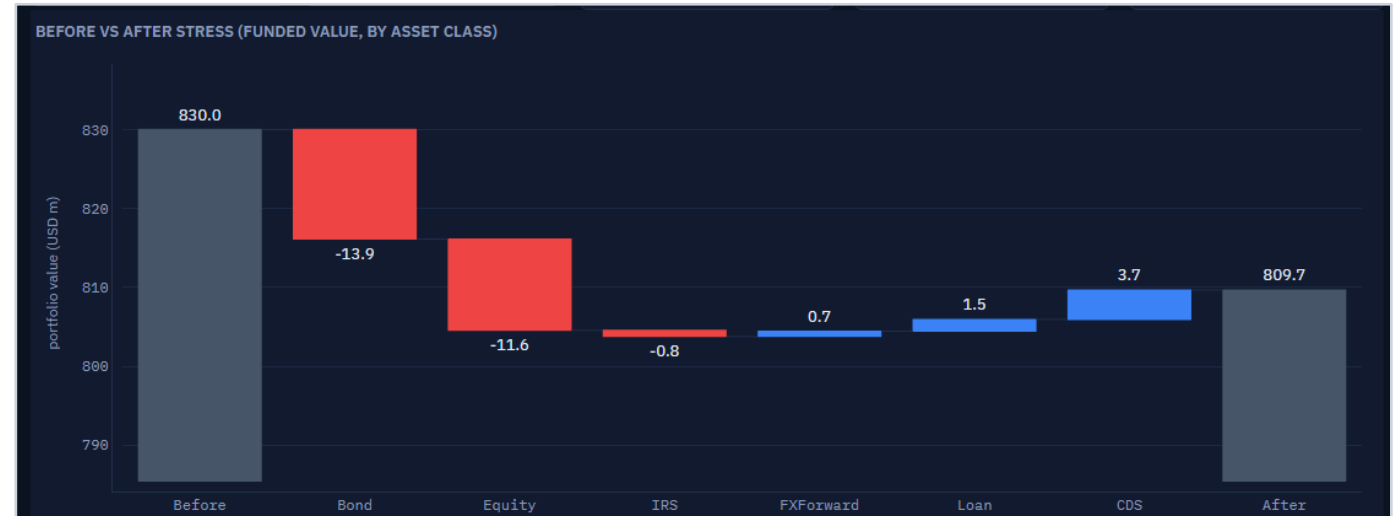
● **0.41% GREEN** Tata Motors downgrade

● **1.32% AMBER** Invasion (1 source)

● **2.45% RED** Corroborated (2 sources)

REAL history: 1,325 captured documents → 1,325 signals → 54 simulated stress runs

Simplified, illustrative hackathon stress model. Not a production or regulatory risk model.



Before → after, corroborated geopolitical shock · Illustrative model, synthetic portfolio

Domain impact and business value

Decision support for credit and portfolio risk

Credit analysts

see which held names need attention and the rule that fired

Risk managers

get an audited, simulated portfolio view when news breaks

Portfolio managers

see linked exposure and test their own shocks instantly

WATCH-NEGATIVE		MONITOR		STABLE		SIGNALS IN WINDOW				
3		6		14		51				
held issuers needing attention		held issuers with negative news		no negative signal of note		on held issuers - last 24 h (event time)				
#	STATUS	ISSUER - RULE	RTG	EXPOSURE	PROPAG.	SIG.	WORST	EVENTS - SOURCES	24 H IMPACT	
1	WATCH	Tata Motors Ltd. (TATAMOTORS.NS) negative signal with impact 8.7 == 7 (sentiment -0.90)	BB	\$41.0m 4,94%	\$23.4m	2 (1)	8.7	Earnings +1, Credit Event +1 google_news		Explain +
2	WATCH	JPMorgan Chase & Co. (JPM) negative signal with impact 7.7 == 7 (sentiment -0.93)	A	\$43.4m 5,23%	\$34.9m	17 (7)	7.7	Other +13, Earnings +4 google_news, mastodon		Explain +
3	WATCH	Tesla Inc. (TSLA) 3 negative signals with impact == 5	BBB	\$40.6m 4,89%	\$23.5m	12 (3)	6.7	Other +9, Earnings +3 google_news		Explain +
4	MONITOR	Microsoft Corp. (MSFT) negative signal with impact 5.2 == 4	AAA-AA	\$24.0m 2,99%	\$18.6m	11 (1)	6.3	Other +9, Earnings +2 google_news		Explain +
5	MONITOR	Nvidia Corp. (NVDA) negative signal with impact 5.2 == 4	AAA-AA	\$13.4m 1,62%	\$62.7m	5 (1)	6.2	Other +2, Supply Chain +1, Earnings +1, Product Launch +1 google_news, mastodon		Explain +
6	MONITOR	Bank of America Corp. 22 linked to US-JPM (peer_of, weight 0.30) WATCH-NEGATIVE	A	\$48.3m 5,82%	\$22.0m	0 (0)	-	-	no signals	

Early-warning watchlist on real news

Credit brief - Tata Motors Ltd. (TATAMOTORS.NS)

Consumer Discretionary - IN - rating bucket BB (illustrative) - as of 2026-10-06T16:13 UTC - WATCH-NEGATIVE

Tata Motors Ltd. is WATCH-NEGATIVE on the early-warning watchlist (2 signals in the 24 h before 2026-10-06T16:13 UTC): negative signal with impact 8.7 >= 7 (sentiment -0.90).

We hold \$41.0m (4.94% of the funded book) across 3 positions, rating bucket BB. Linked issuers add \$23.4m of decay-weighted exposure (2.81%), 7.76% in total.

Under the issuer-only scenario "Idiosyncratic credit event (single issuer)", these positions would change by -\$5.3m, with CDS hedges offsetting \$1.9m (net -\$3.4m): 0.41% of the book.

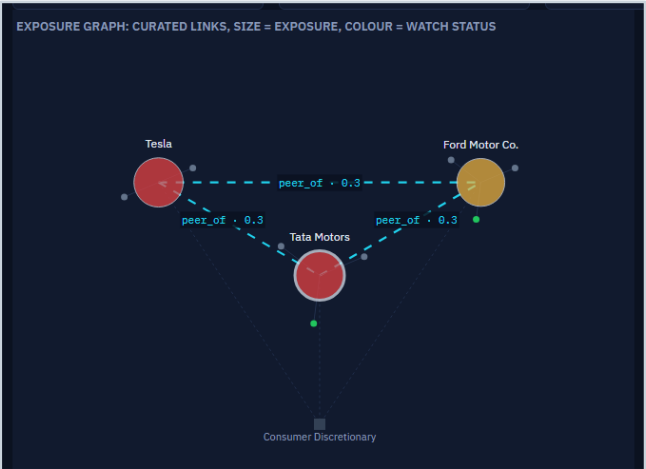
In the latest systemic run (Geopolitical shock (severe), 2026-10-06T16:13 UTC) this issuer's positions moved -\$1.7m; the book lost 2.45% (RED).

LINKED ISSUERS (CURATED)
US-TSLA (peer_of x0.3: \$12.2m), US-F (peer_of x0.3: \$11.2m)

LAST 3 SIGNALS

event time	impact	level	event	sent.	provenance	reason
2026-10-06T16:13	8.7	Critical	Credit Event	-0.90	SYNTHETIC	Impact 8.7 (Critical) driven mainly by Credit Event severity (E = 0.90) and strongly negative sentiment (s = -0.90).
2026-10-	5.6	Medium	Earnings	-0.06	CACHED_REAL	Impact 5.6 (Medium) driven mainly by Earnings severity (E = 0.55) and portfolio exposure to

One-click credit brief (template, no LLM; PDF)



Propagation over curated links

Limitations and future work

Limitations

- Impact weights are uncalibrated priors
- Keyword rules still misfire on real data
- Headline-only news text
- Small AI-labelled evaluation set
- Illustrative stress model, synthetic portfolio

Future work

- Calibrate impact on market moves
- Licensed full-text news
- Fine-tuned models on public labels (notebook ready)
- Kafka streaming
- Backtesting and full revaluation

What already works

- Provenance on every record
- One pipeline for every input
- Real-data time machine
- Trigger audit trail
- Offline demo on a laptop

An honest, explainable bridge from text to portfolio risk that runs offline on a laptop.